

DIFFERENTIAL COHOMOLOGY SEMINAR 10

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Today we look at the construction of pushforward maps in twisted K-theory due to Carey and Wang [CW08].

1. THE CLASSICAL CASE

Let us do a quick review of the classical case. Given a map $f: X \rightarrow Y$ and induced map on K-theory $f^*: K(Y) \rightarrow K(X)$, we would like to construct a pushforward map $f_!: K(X) \rightarrow K(Y)$. In the classical case, this requires $w_2(X) = f^*w_2(Y)$. This precisely corresponds to choosing a Spin^c -structure on the virtual bundle $TX - f^*TY$. This is the necessary condition to define this map.

2. REVIEWING TWISTED K-THEORY

Before we proceed to generalize the pushforward map to twisted K-theory, we first review the definition of twisted K-theory. Recall that

$$H^3(M, \mathbb{Z}) \cong PU(\mathcal{H}) - \mathcal{B}dl(M) / \sim$$

where $\mathcal{H}$ is a separable Hilbert space and $PU(\mathcal{H})$ is the projective unitary group. Choosing P a principal $PU(\mathcal{H})$ -bundle over M representing a class in $H^3(M, \mathbb{Z})$, we can form the associated bundle of Fredholm operators $\mathcal{F}red(P) := P \times_{PU(\mathcal{H})} \mathcal{F}red(\mathcal{H})$. Then the 0-th twisted K-theory is defined as

$$K^0(M, P) := \pi_0(\text{Map}(M, P \times_{PU(\mathcal{H})} \mathcal{F}red(\mathcal{H}))) \cong \pi_0(\text{Map}(P, \mathcal{F}red(\mathcal{H})))_{PU(\mathcal{H})}.$$

More generally

$$K^n(M, P) := \pi_0(\text{Map}(P, \Omega^n \mathcal{F}red(\mathcal{H})))_{PU(\mathcal{H})}.$$

In the coming sections we consider the pushforward map in two cases: the torsion case and the non-torsion case.

3. UNDERSTANDING THE TORSION CASE

Let us consider the case when the bundle is torsion. Then P is a principal $PU(n)$ -bundle, using $U(n) \hookrightarrow U(\mathcal{H})$. In this case we can obtain a bundle gerbe $\mathcal{L}_P$, which has the following form

$$\begin{array}{ccccc} U(n) & \longleftarrow & \Gamma_P & & E \\ \downarrow & & \downarrow & & \downarrow \\ PU(n) & \longleftarrow & P \times_M P & \xrightarrow{\quad} & P \\ & & & & \downarrow \\ & & & & M \end{array}$$

This structure gives us a map

$$\phi: (\Gamma_P|_{p_1, p_2} \times_{U(1)} \mathbb{C}) \otimes E_{p_2} \rightarrow E_{p_1}$$

meaning (E, ϕ) is a $\mathcal{L}_P$ -twisted vector bundle. We now can prove the following theorem.

Theorem 3.1. *Let P be a principal $PU(n)$ -bundle. Then, we have an isomorphism $K(\mathcal{L} - \text{Mod}) \cong K_{U(n), \text{scal}}(P)$.*

Here $K_{U(n), \text{scal}}(P)$ is the $U(n)$ -equivariant K-theory of P with scalar action of the center $U(1) \subset U(n)$. The proof proceeds by mapping a pair (E, ϕ) to the data $(U(n) \times E \rightarrow E)$ that maps $(A, v) \mapsto \phi(A, v)$.

We now have the following second theorem, relating twisted K-theory to twisted vector bundles.

Theorem 3.2. *Let P be a principal $PU(n)$ -bundle. Then we have an isomorphism*

$$K^0(M, P) \cong K(\mathcal{L}_P - \text{Mod}).$$

Here we are implicitly using the $PU(\mathcal{H})$ -bundle $\tilde{P} = P \times_{PU(n)} PU(\mathcal{H})$, meaning

$$K^0(M, P) = K^0(M, \tilde{P}) = \pi_0(\text{Map}(\tilde{P}, \mathcal{F}\text{red}(\mathcal{H})))_{PU(\mathcal{H})}.$$

Now we also have $\mathcal{F}\text{red}(\mathcal{H}) \cong \mathcal{F}\text{red}_{U(n)-cts}(\mathcal{H})$, meaning it is the Fredholm operators with continuous $U(n)$ -action.

Now before we proceed, recall that we have an equivariant version of Atiyah-Jänich theorem (which holds if certain conditions are satisfied), meaning

$$[M, \mathcal{F}\text{red}_{G-cts}(\mathcal{H})]_G \cong K^0(\text{Vect}_G(M)).$$

We can now restrict this bijection to a bijection of subsets

$$[M, \mathcal{F}\text{red}_{U(n)-cts}(\mathcal{H}^{\text{scal}})]_G \cong K^0(\text{Vect}_{U(n), \text{scal}}(M)),$$

where $\mathcal{H}^{\text{scal}}$ is the Hilbert space with scalar action of the center $U(1) \subset U(n)$.

The construction proceeds now by replacing $\mathcal{H}$ with $\mathcal{H}^{\text{scal}}$ in the definition of twisted K-theory, meaning we go back to the first step and do

$$K^0(M, P) = K^0(M, \tilde{P}) = \pi_0(\text{Map}(\tilde{P}, \mathcal{F}\text{red}(\mathcal{H}^{\text{scal}})))_{PU(\mathcal{H})}.$$

which via the bijection above is isomorphic to $K_{U(n), \text{scal}}(P)$, which by the previous theorem is isomorphic to $K(\mathcal{L}_P - \text{Mod})$.

Remark 3.3. Due to recent work, we know that if P is torsion, then $\mathcal{L}_P \cong \mathcal{A}_P = P \times_{PU(n)} \mathbb{C}^{n \times n}$, which is an isomorphism of 2-vector bundles, and induces an isomorphism $\mathcal{L}_P - \text{Mod} \cong \mathcal{A}_P - \text{Mod}$.

4. THOM ISOMORPHISM IN THE TORSION CASE

Let $V \rightarrow M$ be a $\mathbb{R}$ -vector bundle. Then we get $\text{Tr}(V) \rightarrow M$, the associated $SO(n)$ -bundle, which comes with a sub-bundle with a $\text{Spin}^c(n)$ -structure. This gives us a bijection $\mathcal{L}_{\text{Tr}(V)} \cong CL(V)$, where $CL(V)$ is the complex Clifford bundle of V .

Here we can use the result by Karoubi.

Theorem 4.1. *We have an isomorphism $K(CL(V) - \text{Mod}) \cong K(\text{Th}(V))$.*

We now want to generalize this to more general twists, which should be the following result.

Theorem 4.2. *There is an isomorphism*

$$K^0(M, P + W_3(V)) \cong K^0(\text{Th}(V), \pi^*P).$$

5. NON-TORSION CASE

We now aim to generalize these results to the non-torsion case. Concretely, we want prove an analogue of the following theorem:

$$K(\mathcal{L}_P - \text{Mod}) \cong K_{U(n), \text{scal}}(P)$$

We can try to reproduce the same diagram

$$\begin{array}{ccc} \Gamma_P & & E \\ \downarrow & & \downarrow \\ P \times_M P & \xrightarrow{\quad} & P \\ & & \downarrow \\ & & M \end{array},$$

but we cannot compare to $U(n)$ anymore, since P is not a $PU(n)$ -bundle. So, we do not get the original result, but rather a restricted version.

Theorem 5.1. $K^0(M, P) \cong K(\mathcal{L}_P - \text{Mod}^{U_2})$.

Here $U_2 \subseteq U(\mathcal{H})$ is the group of unitary objects that differ from the identity by a Hilbert-Schmidt operator, and $\mathcal{L}_P - \text{Mod}^{U_2}$ is the category of $\mathcal{L}_P$ -twisted vector bundles whose transition functions take values in U_2 .

Here U^2 can also be described via colimits, as $U^2 \cong \text{colim}_{n \rightarrow \infty} U(n)$. So this result can be interpreted as the fact that filtered colimits preserve the original theorem.

Now we analogously have an isomorphism

$$\mathcal{L}_P - \text{Mod}^{U_2} \cong \mathcal{A}_p - \text{Mod}^{U_2}.$$

Remark 5.2. Similar to above, we anticipate an isomorphism of 2-vector bundles $\mathcal{L}_P \cong P \times_{PU(\mathcal{H})} \text{HS}(\mathcal{H})$, that induces the isomorphism above as a special case. This would require a suitable Morita category for these Hilbert-Schmidt operators, which is currently unknown.

6. TWISTED PUSHFORWARD

As a last step we can deduce the Thom isomorphism in this non-torsion case. Let $f: X \rightarrow Y$. Then we have $W_3(f) := W_3(TX \oplus f^*TY)$. This gives us a pushforward map

$$f_!: K^*(X, W_3(f) + f^*P) \rightarrow K^{*(+(\dim X - \dim Y) \bmod 2)}(Y, P),$$

for $P \rightarrow Y$ some $PU(\mathcal{H})$ -bundle.

The construction of this map proceeds as follows: We can factor $X \rightarrow Y$ into $X \hookrightarrow Y \times S^n \rightarrow Y$. The first step uses the Thom isomorphism, and the second step uses C^* -algebra techniques.

Remark 6.1. If Y is trivial, then this construction gives us a version of an index.

7. D-BRANES

There is an application of these methods to physics. Given a $U(n)$ -bundle $\mathcal{G}$ over X with sub objects Q , such that the restriction along Q has a $\mathcal{G}|_Q$ -module $\mathcal{E}$.

It is claimed by Witten that these d-branes are classified by some version of K-theory. This can be made precise as follows (with some adjustments). Given $\mathcal{E}$ and $\mathcal{G}|_Q \otimes \mathcal{L}_f$ -module we get an element in $K^0(Q, \mathcal{G}|_Q \otimes \mathcal{L}_f)$. Using the pushforward map, we have a map

$$K^0(Q, \mathcal{G}|_Q \otimes \mathcal{L}_f) \rightarrow K^*(X, \mathcal{G})$$

giving us an element in $K^*(X, \mathcal{G})$, as desired.

We can in particular consider the case when $X = \mathcal{G}$, which is known as the WZW-model in physics. In this case, Q are conjugacy classes C_g of some element g in the group and for $f: C_g \rightarrow \mathcal{G}$, we have

$$W_3(f) = (f^* \mathcal{G}_{bas})^{\check{c}}$$

This is a non-trivial observation. Indeed comparing with the Freed-Hopkins-Teleman approach [FHT11], we see that

$$K_{\mathcal{G}}(\mathcal{G}, \mathcal{G}_{bas}^k) \cong \text{Rep}^{k+\check{c}}(L\mathcal{G}),$$

meaning this $\check{c}$ Coxeter shift appears naturally in the representation theory of the loop group.

Hence this approach suggests a possible geometric interpretation of these constructions.

REFERENCES

- [CW08] Alan L. Carey and Bai-Ling Wang. Thom isomorphism and push-forward map in twisted K -theory. *J. K-Theory*, 1(2):357–393, 2008.
- [FHT11] Daniel S. Freed, Michael J. Hopkins, and Constantin Teleman. Loop groups and twisted K -theory I. *J. Topol.*, 4(4):737–798, 2011.